PPGBG (v1.0) Release Overview

Dataset name: PPGBG
Version: v1.0
Application contact: PPGBGa@163.com
Access condition: Non-commercial academic research under the PPGBG Data Use Agreement.

PPGBG is a longitudinal, subject-independent benchmark dataset for non-invasive blood glucose estimation from fingertip photoplethysmography (PPG), with synchronized cuff blood pressure measurements and clinical metadata.

Scale item	Value
Subjects	183
Clinical encounters	528
Valid 40 s PPG acquisitions	523
Validated 4 s PPG windows	5,224
Signal format	Fingertip PPG, 1000 Hz, 4 s windows, 4000 samples per window

Release Structure

```
PPGBG_v1.0_release/
|-- data/
|   |-- signals.npy
|   |-- windows.csv
|   |-- encounters.csv
|   `-- subjects.csv
|-- documents/
|   |-- en/
|   |   |-- README.pdf
|   |   |-- DATASET_DOCUMENTATION.pdf
|   |   `-- DATA_USE_AGREEMENT.pdf
|   `-- zh/
|       |-- README_zh.pdf
|       |-- DATASET_DOCUMENTATION_zh.pdf
|       `-- DATA_USE_AGREEMENT_zh.pdf
`-- SHA256SUMS.txt
```

Data Files

File	Description
`data/signals.npy`	PPG windows, `float32`, shape `(5224, 4000)`; row-aligned to `windows.csv`.
`data/windows.csv`	Per-window labels, including BG, BP, HR, demographics, comorbidities, acute-condition indicators, and smoking status.

`data/encounters.csv`	Per-encounter clinical summaries.
`data/subjects.csv`	Per-subject demographics, comorbidities, analysis groups, and window counts.

Access and Citation

Access requests are submitted by completing `documents/en/DATA_USE_AGREEMENT.pdf` or `documents/zh/DATA_USE_AGREEMENT_zh.pdf` and sending the signed document to **PPGBGa@163.com**.

Works using PPGBG must cite the associated publication:

> Zhang, Z., Li, J., Wang, Y., Yang, K., and Lei, T. (2026). Dynamic Incremental Learning for Non-invasive Blood Glucose Estimation from Wearable Physiological Signals. Manuscript under review. Preprint DOI: https://doi.org/10.5281/zenodo.21202402

Quick Start

```
import numpy as np
import pandas as pd

X = np.load("data/signals.npy")       # (5224, 4000), float32
W = pd.read_csv("data/windows.csv")   # row i describes X[i]
y = W["bg"].to_numpy()                # blood glucose target, mmol/L
```

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